深度學習
Deep Learning

圖神經網路
Graph Neural Network (GNN)

Instructor: 林英嘉 (Ying-Jia Lin)
2025/12/03

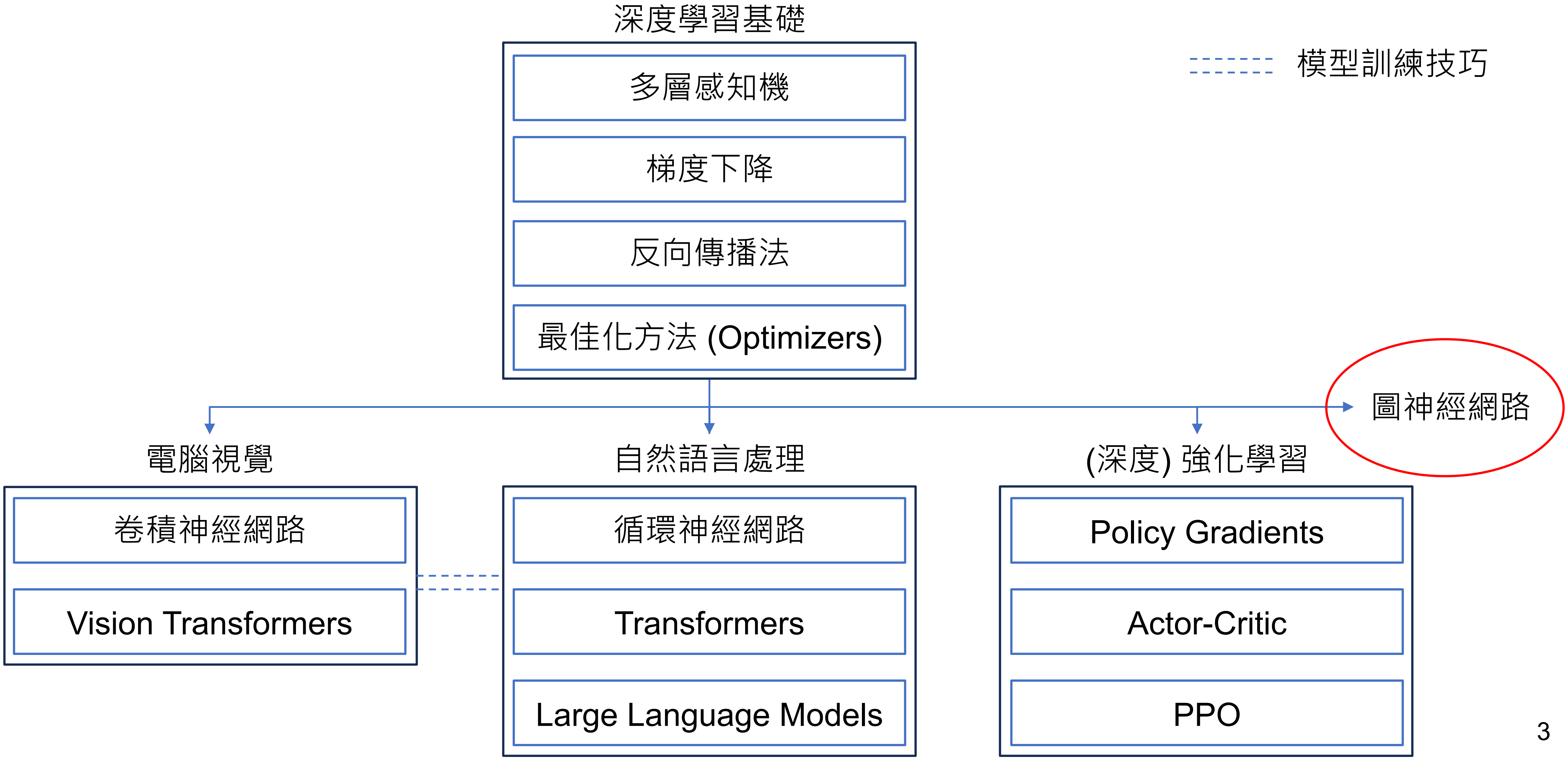
Course GitHub

Slido # DL1203

Outline

- 圖資料結構介紹 – 20 min
- Graph Neural Networks
  - GCN (Graph Convolutional Network) – 20 min
  - GAT (Graph Attention Network) – 15 min
- Code – 30 min
- Project presentations – 40 min
- Quiz – 15 min

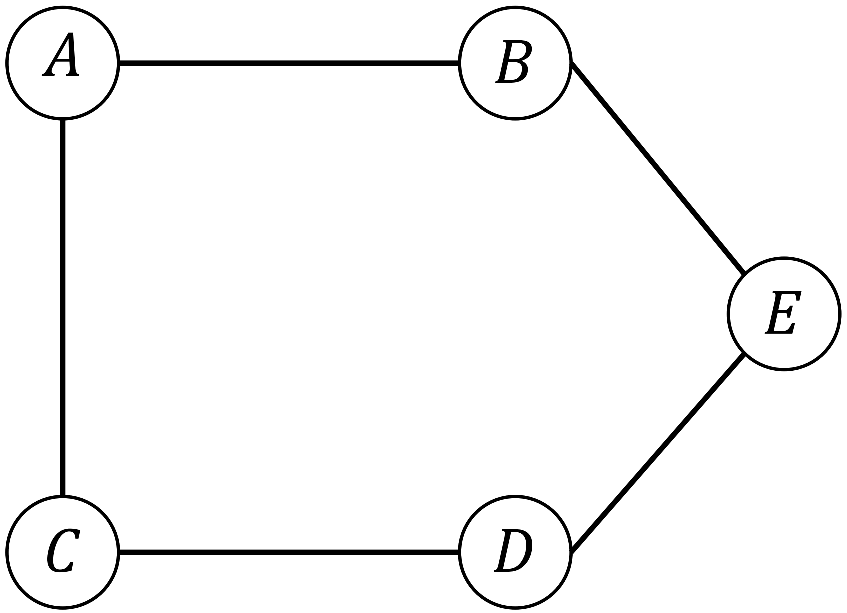
深度學習課程地圖



深度學習基礎
多層感知機
梯度下降
反向傳播法
最佳化方法 (Optimizers)
模型訓練技巧
圖神經網路
電腦視覺
卷積神經網路
Vision Transformers
自然語言處理
循環神經網路
Transformers
Large Language Models
(深度) 強化學習
Policy Gradients
Actor-Critic
PPO

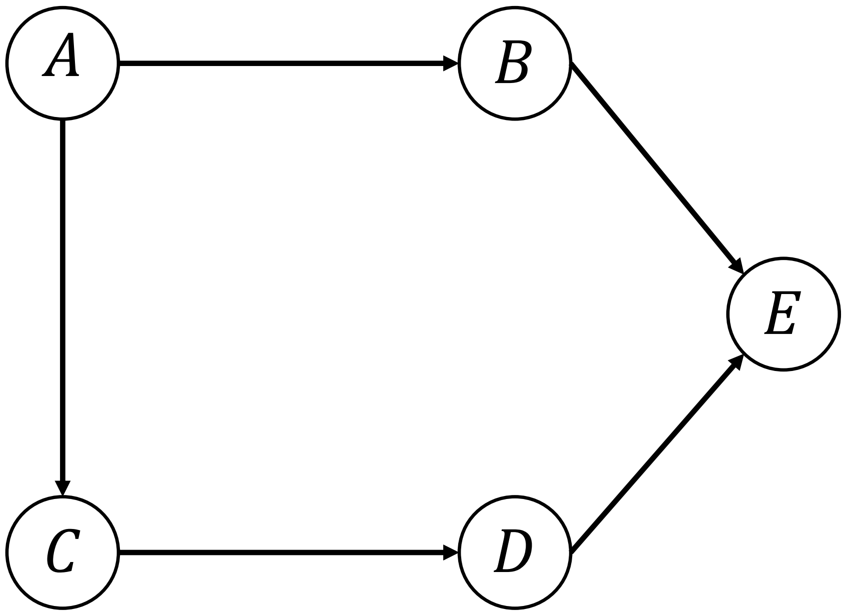
Graph examples

Node (節點) Edge (邊)



cyclic

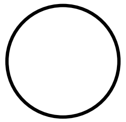
Undirected graph (無向圖)

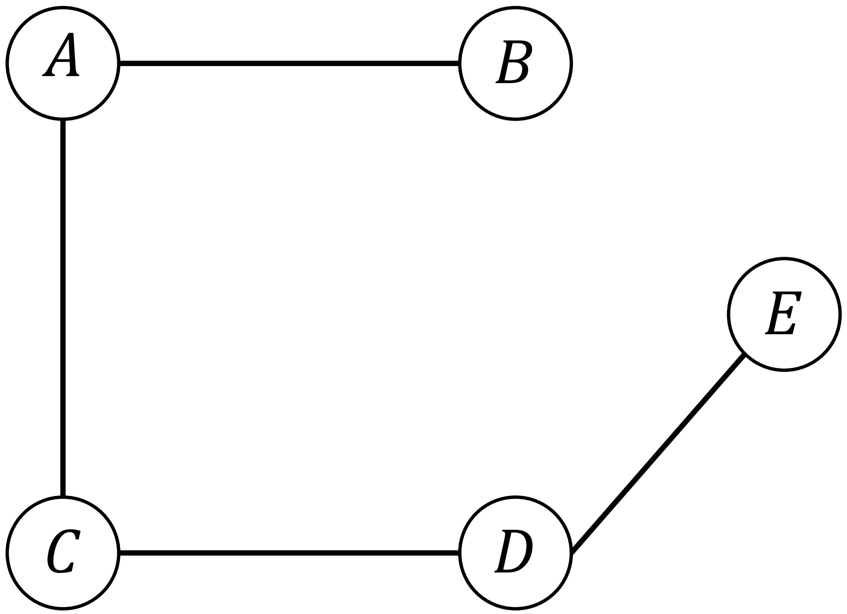


acyclic

Directed graph (有向圖)

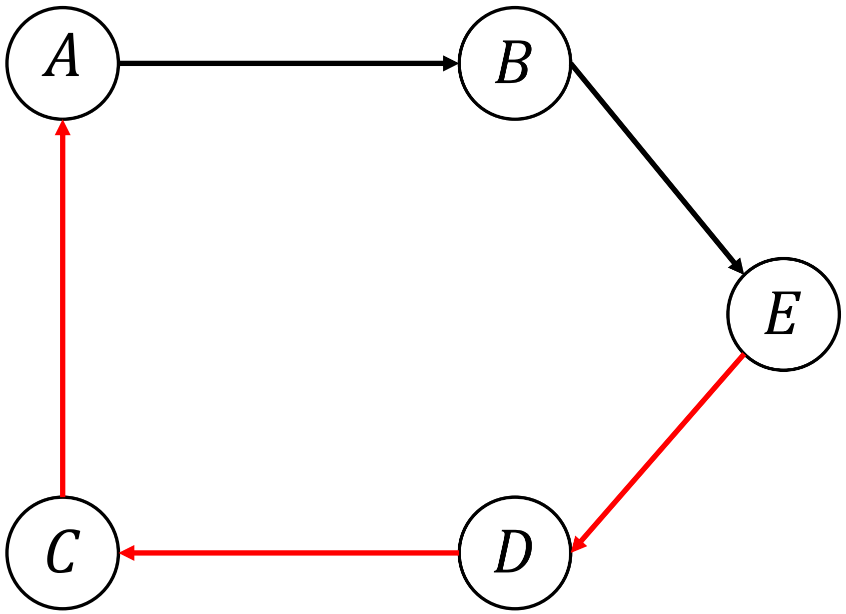
Graph examples

 Node (節點)  Edge (邊)



acyclic

Undirected graph (無向圖)



cyclic

Directed graph (有向圖)

Graph 的應用例子 (推薦系統)

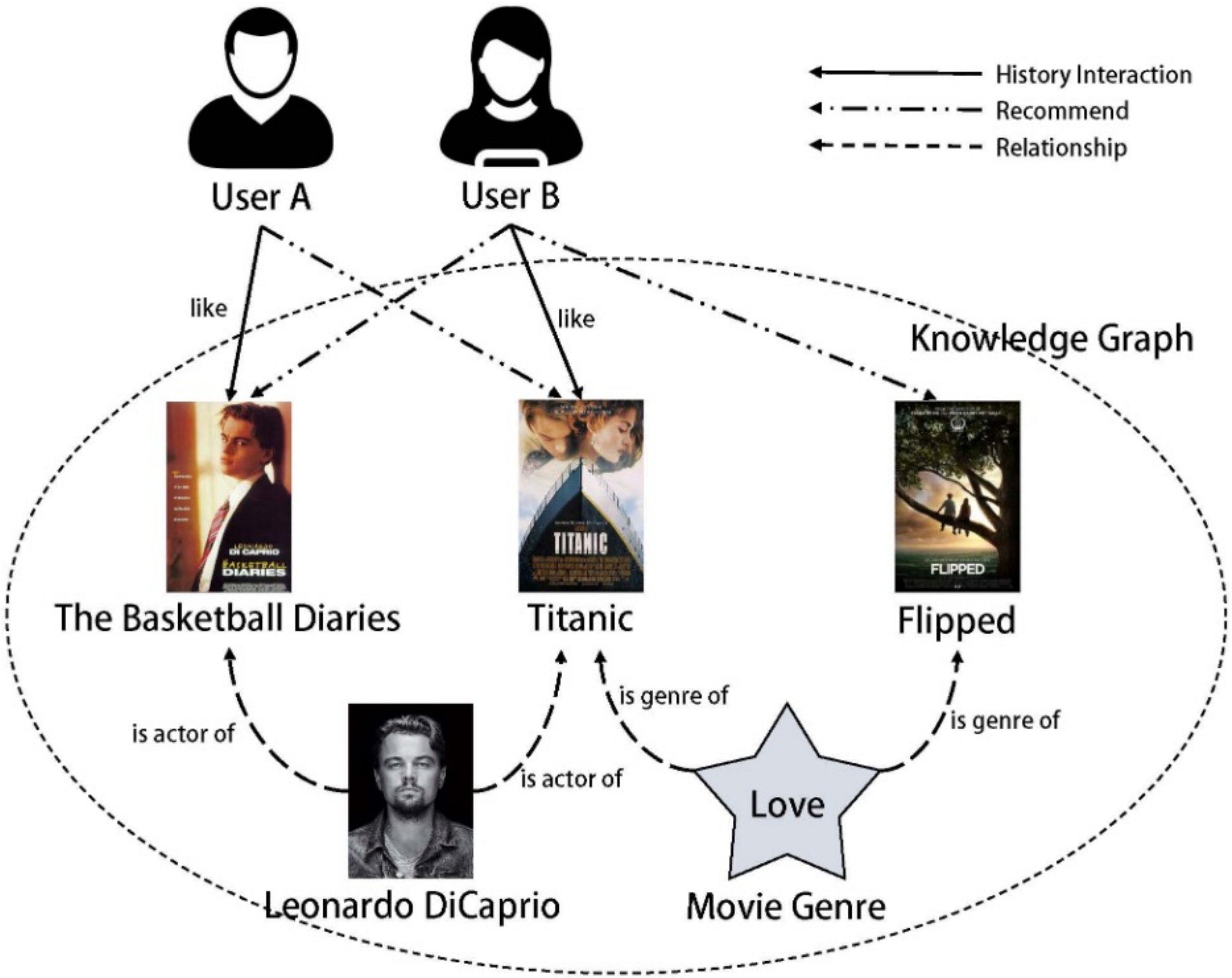


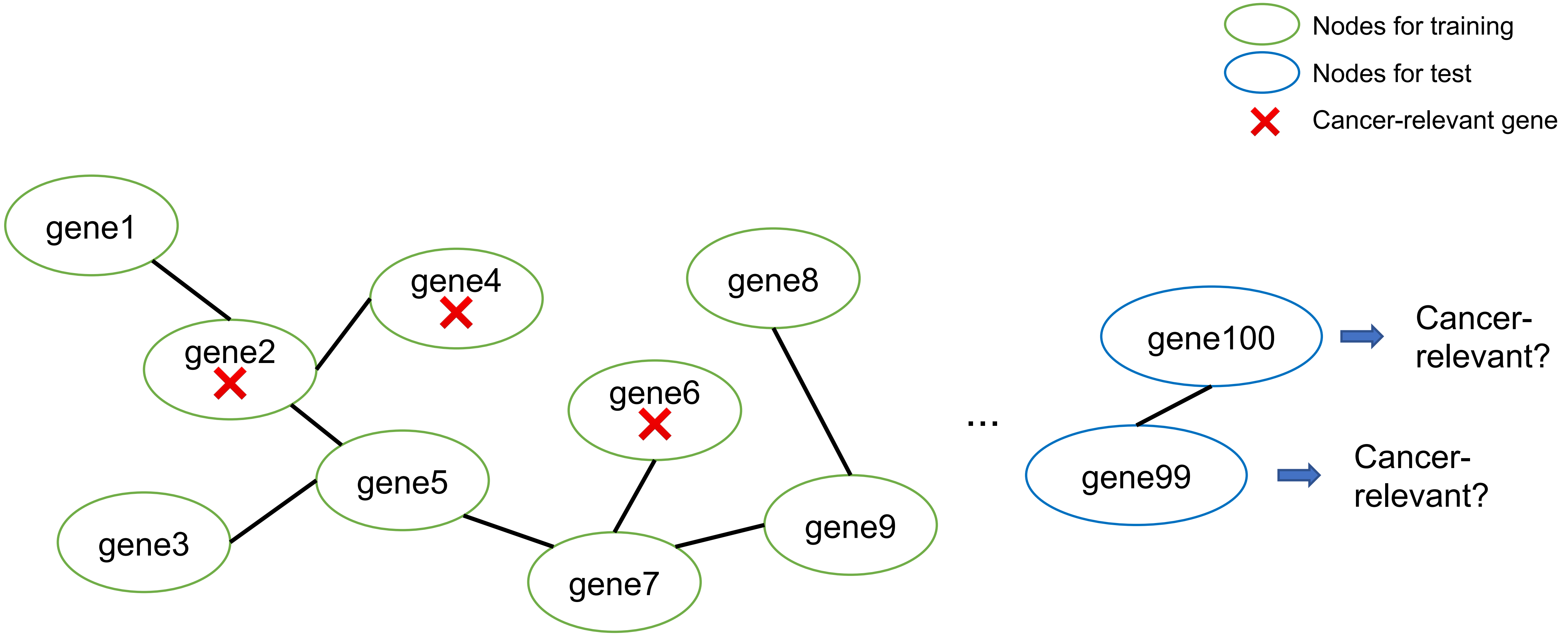
Figure source: Huang, Xiaoli, Junjie Wang, and Junying Cui. "A Personalized Collaborative Filtering Recommendation System Based on Bi-Graph Embedding and Causal Reasoning." Entropy 26.5 (2024): 371.

Graph 的應用例子 (基因關聯)

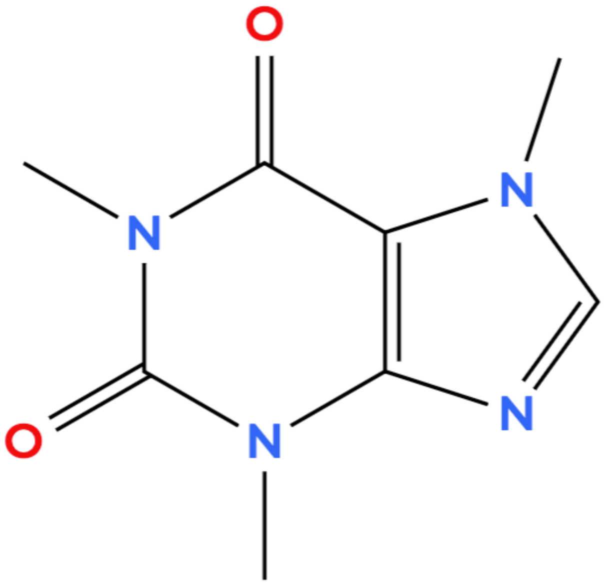
Figure 2. - A visual display of Kyoto Encyclopedia of Genes and Genomes pathways in 6 arsenic trioxide target genes (AKT1, CCND1, CDKN2A, IKBKB, MAPK1 and MAPK3) associated with pancreatic cancer.

Figure source: Zhou, Cong-Ya, et al. "Evaluation of the target genes of arsenic trioxide in pancreatic cancer by bioinformatics analysis." Oncology Letters 18.5 (2019): 5163-5172.

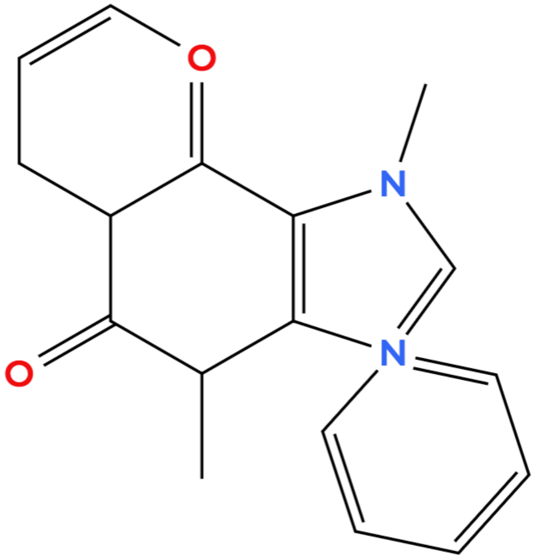
圖常見任務 (1)：Node Classification



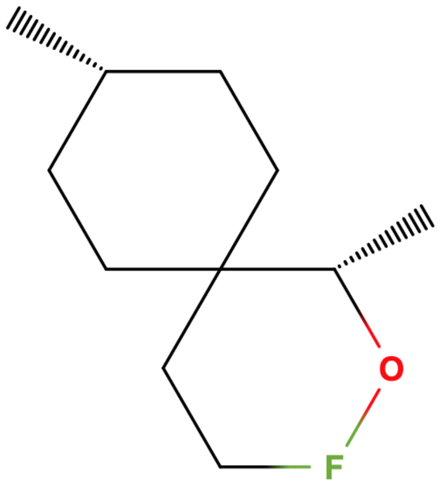
圖常見任務 (2)：Graph Classification



Toxic



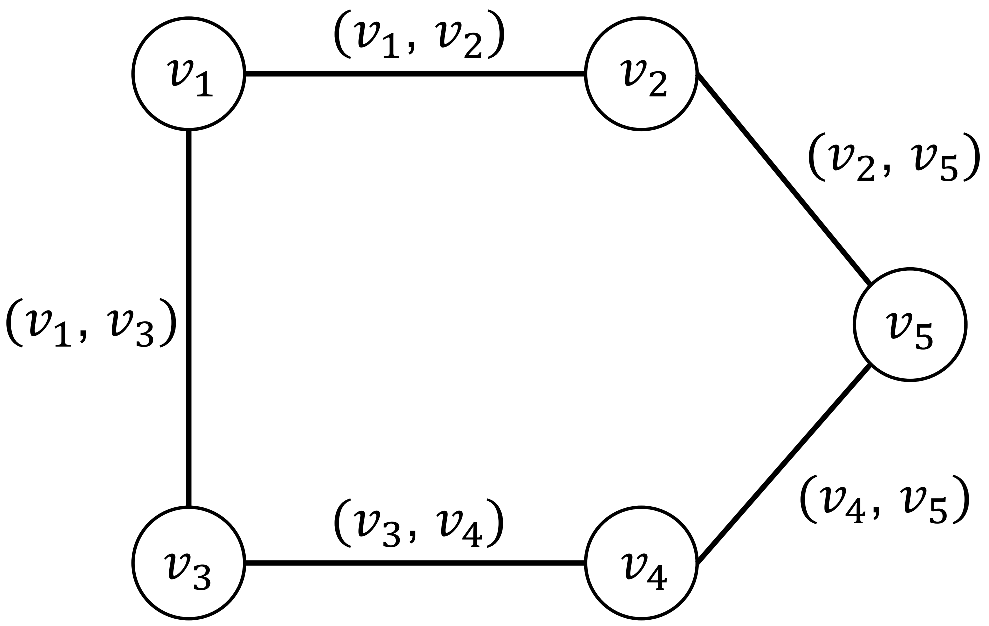
Not Toxic



?

Hu, Weihua, et al. "Open graph benchmark: Datasets for machine learning on graphs." Advances in neural information processing systems 33 (2020): 22118-22133.

Graph Terminology



Undirected graph (無向圖)

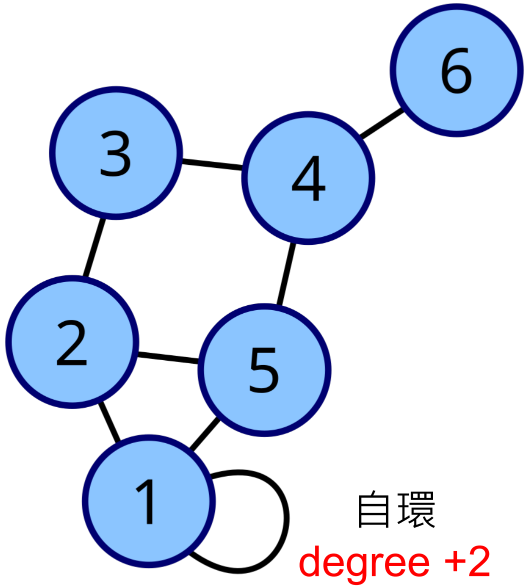
一張圖 G 寫作 $G = (V, E)$，

其中：

- $V = \{v_1, v_2, \ldots, v_n\}$
- $E \subseteq V \times V$，如 (v_i, v_j)， (i, j)為 G 中任意相鄰的兩個節點 (node)
  - i 通常代表起點；
  - j 通常代表終點

Adjacency Matrix and Degree Matrix

Rows 代表 node i ; Columns 代表 nodes j

Vertex labeled graph	**Degree matrix** **(度矩陣)**	**Adjacency matrix** **(鄰接矩陣)**
	$D=\begin{pmatrix}4&0&0&0&0&0\\0&3&0&0&0&0\\0&0&2&0&0&0\\0&0&0&3&0&0\\0&0&0&0&3&0\\0&0&0&0&0&1\end{pmatrix}$ 對角矩陣	$A=\begin{pmatrix}1&1&0&0&1&0\\1&0&1&0&1&0\\0&1&0&1&0&0\\0&0&1&0&1&1\\1&1&0&1&0&0\\0&0&0&1&0&0\end{pmatrix}$

Graph Neural Networks (GNN)

GNN 核心步驟

以 Node classification 為例

Step1: 聚合鄰居節點資訊 AGGREGATE

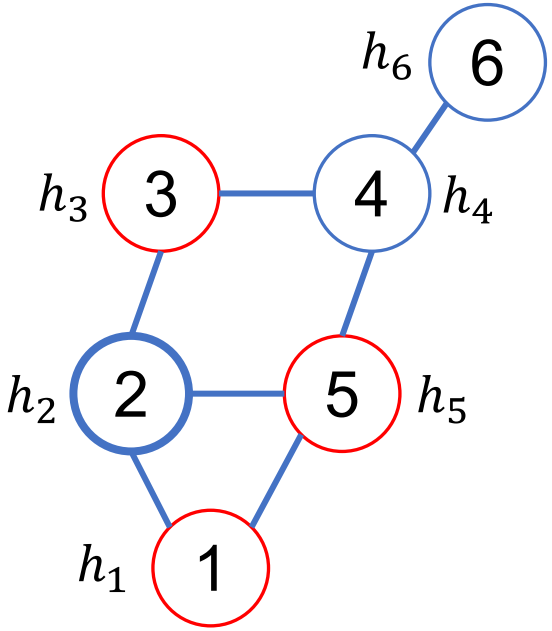
↓

Step2: 更新自身節點資訊 UPDATE

GNN 的核心步驟 (1)：聚合 (Aggregate)

💡 Aggregate: 收集鄰居資訊

v_i v_j



$$\mathcal{N}(v_2) = \{v_1, v_3, v_5\}$$

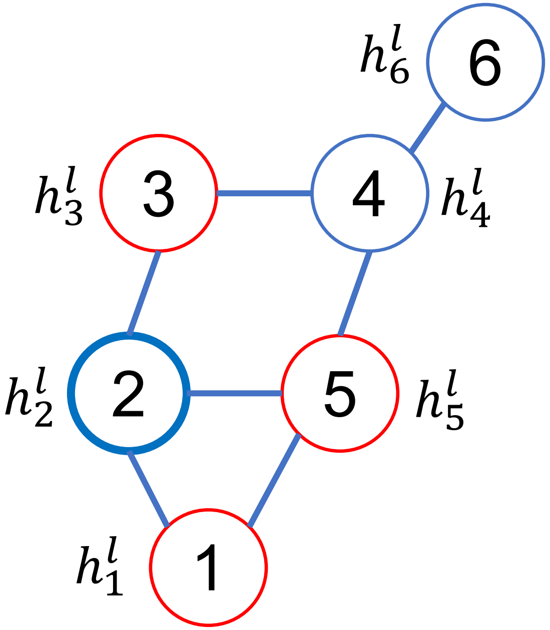
$$m_i = \mathrm{AGGREGATE}(\{h_j | v_j \in \mathcal{N}(v_i)\})$$

h_n: 節點 v_n 的特徵表示

GNN 的核心步驟 (1)：聚合 (Aggregate)

v_i v_j

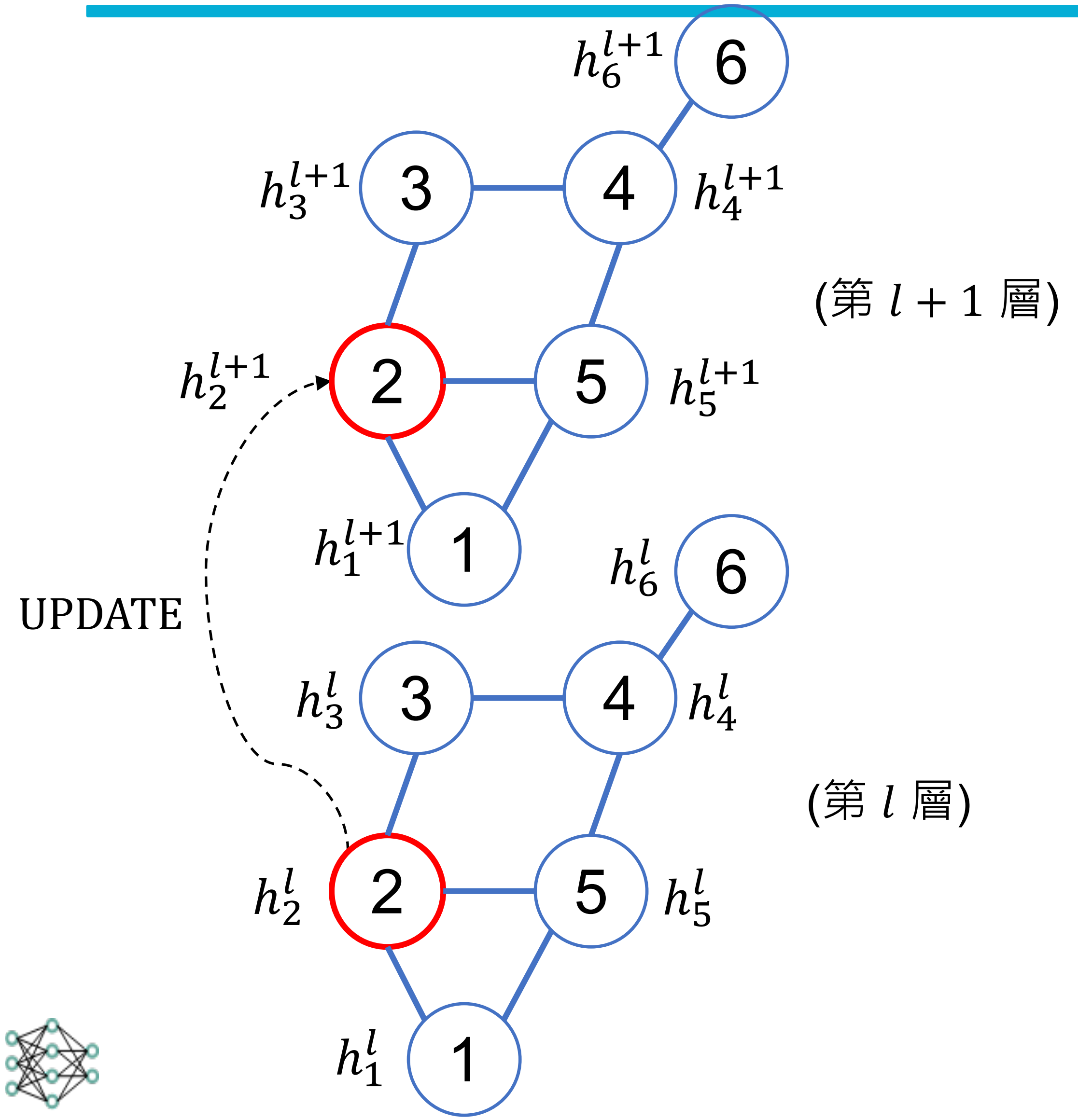
Aggregate: 收集鄰居資訊



$$m_i^l = \text{AGGREGATE}(\{h_j^l | v_j \in \mathcal{N}(v_i)\})$$

h_n^l: **第 l 層中**，節點 v_n 的特徵表示

GNN 的核心步驟 (2)：更新 (Update)



💡Combine: 收集到鄰居訊息後，節點會根據這些資訊更新自己的特徵表示

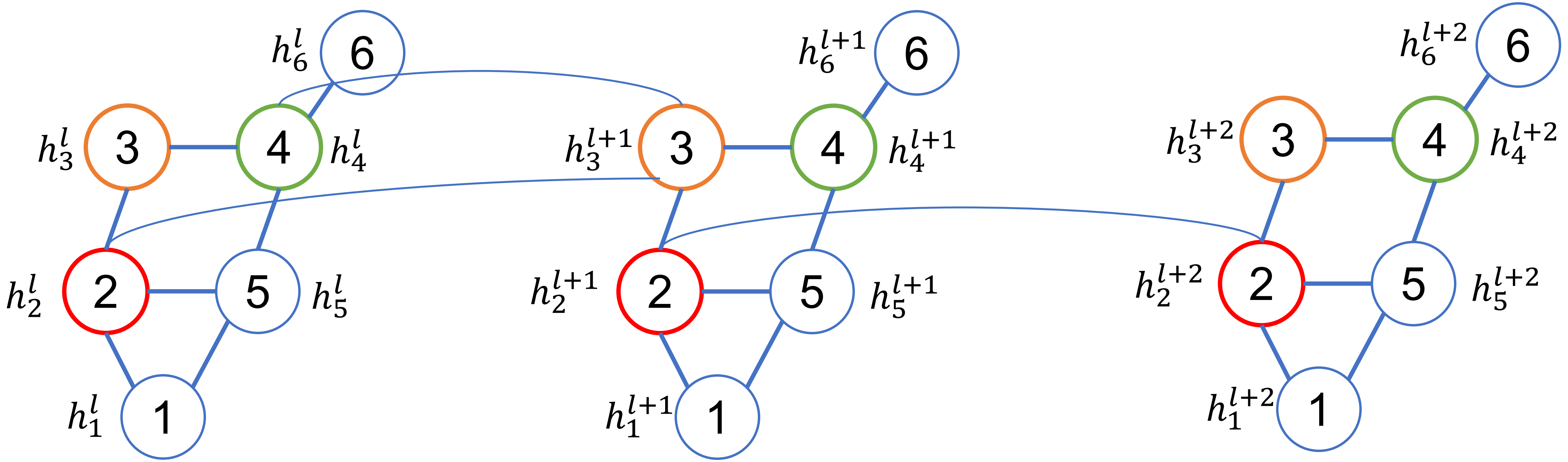
$$h_i^{l+1} = \mathrm{UPDATE}(h_i^l, m_i^l)$$

h_n^l: **第 l 層中**，節點 v_n 的特徵表示

Message Passing

💡一個節點的特徵，應該由它自身的特徵以及它所有鄰居的特徵共同決定。透過多層 (Layers) 的訊息傳遞，節點可以獲取更遠距離的資訊

Message Passing 說明範例 (1)



(第 l 層)

2號node無法接觸到4號node

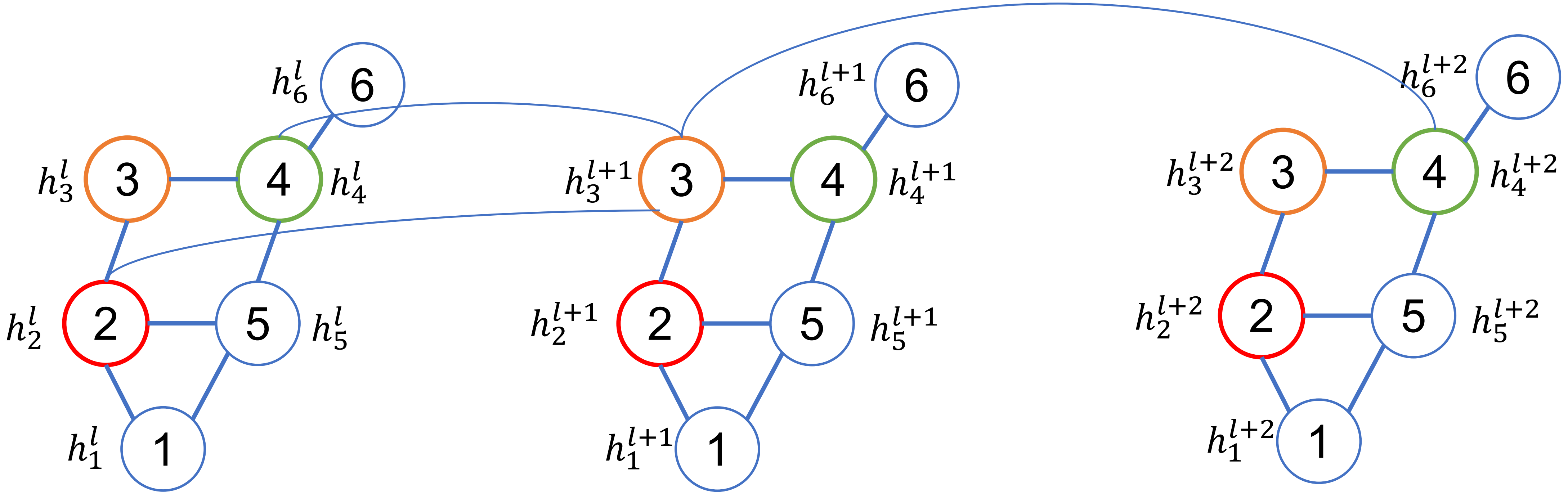
(第 $l+1$ 層)

3號node帶有4號node和2號node的資訊

(第 $l+2$ 層)

2號node帶有4號node的資訊

Message Passing 說明範例 (2)



(第 l 層)

2號node無法接觸
到4號node

(第 $l+1$ 層)

3號node帶有4號node
和2號node的資訊

(第 $l+2$ 層)

4號node帶有2號
node的資訊

Graph Convolutional Networks (GCN)

Thomas N. Kipf, , Max Welling. "Semi-Supervised Classification with Graph Convolutional Networks." International Conference on Learning Representations. 2017.

(Recap) GNN 核心步驟

以 Node classification 為例

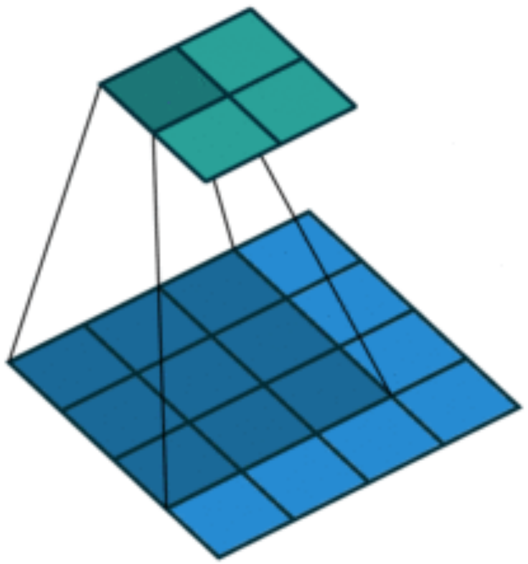
Step1: 聚合鄰居節點資訊 AGGREGATE

↓

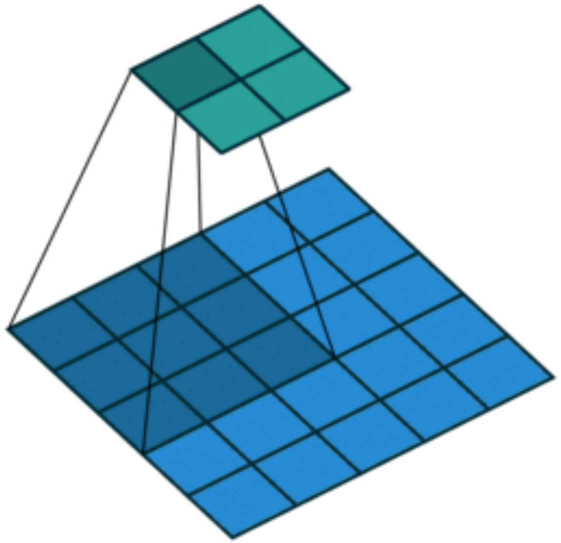
Step2: 更新自身節點資訊 UPDATE

(Recap) Examples of Convolutions

Padding = 0, stride = 1



Padding = 0, stride = 2

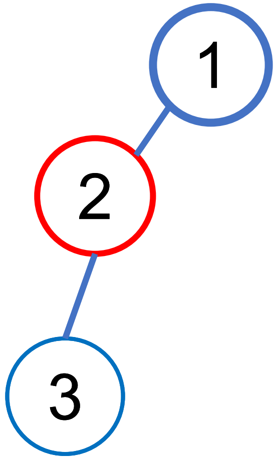


💡卷積 (convolution) 就是讓一個「小的矩陣」(filter) 在影像上滑動，進行局部加權平均 (局部區域的內積)，產生新的影像 (特徵圖，feature map)

Figure source:
https://hannibunny.github.io/mlbook/neuralnetworks/convolutionDemos.html

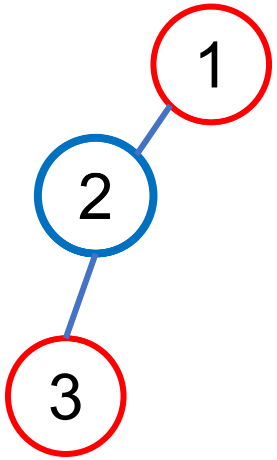
GCN 會先對鄰居收集資訊

Aggregate: 收集鄰居資訊



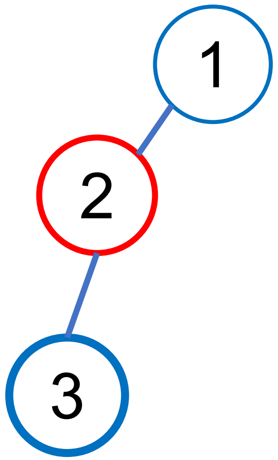
Source node: **1**

Neighbor node: **2**



Source node: **2**

Neighbor node: **1, 3**

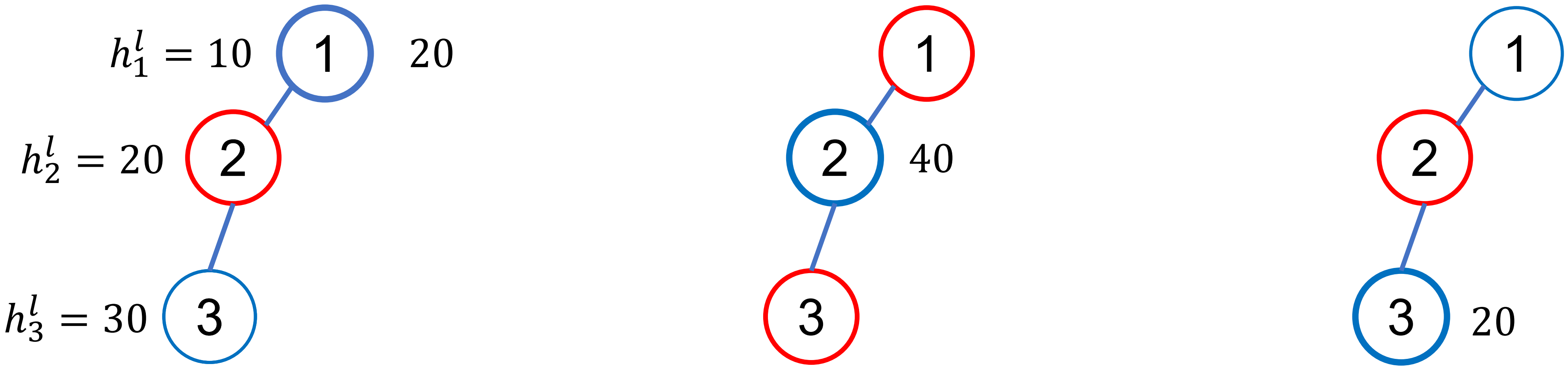


Source node: **3**

Neighbor node: **2**

GCN 如何做 AGGREGATE？

💡GCN 使用相加來收集鄰居資訊



Source node: **1**

Neighbor node: **2**

Source node: **2**

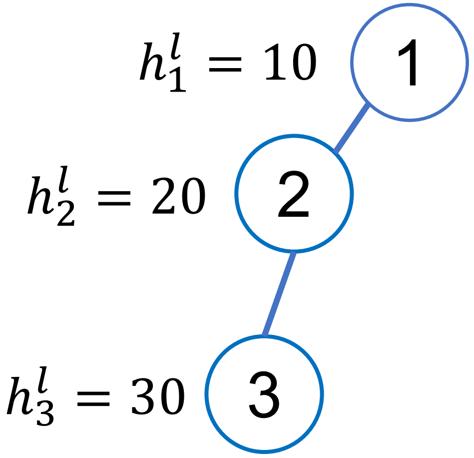
Neighbor node: **1, 3**

Source node: **3**

Neighbor node: **2**

⚠️ 此頁slide為求簡潔，每個node僅使用 1 個feature

實作上 GCN 如何做 AGGREGATE？



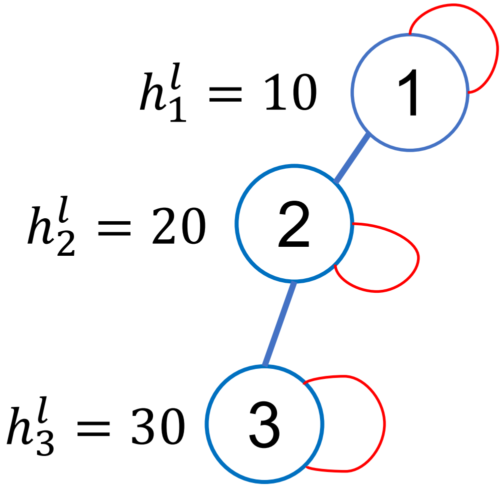
Adjacency matrix
(鄰接矩陣)

$$\tilde{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$H = \begin{pmatrix} 10 \\ 20 \\ 30 \end{pmatrix}$$

$$\tilde{A}H = \begin{pmatrix} 20 \\ 40 \\ 20 \end{pmatrix}$$

實作上 GCN 如何做 AGGREGATE？



$$\tilde{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \xrightarrow{\text{加入自環}} \tilde{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

💡不僅要看鄰居，也要保留自己的資訊。

每個 h 有什麼內容物

```
# 3. 查看節點特徵向量的內容與形狀

print(f"特徵向量的內容: {dataset[0].x[0]}")
print(f"特徵向量的形狀: {dataset[0].x[0].size()}")
```

```
特徵向量的內容: tensor([0., 0., 0.,  ..., 0., 0., 0.])
特徵向量的形狀: torch.Size([1433])
```

https://github.com/mcps5601/CGUDL_2025_fall/blob/main/code/gnn.ipynb

(Recap) GNN 核心步驟

以 Node classification 為例

Step1: 聚合鄰居節點資訊 AGGREGATE

↓

Step2: 更新自身節點資訊 UPDATE

GCN 的 weights (UPDATE)

假設有一張圖的 nodes 有3個，且每個 node 的 feature 數目為10個

GCN_layer1 $H^{(2)} = \tilde{A}H^{(1)}W$

$\tilde{A}$: 3×3

$H^{(1)}$: 3×10

W: $10 \times \text{hidden\_size}$

GCN_layer2 $H^{(3)} = \tilde{A}H^{(2)}W$ → **Node Classification**

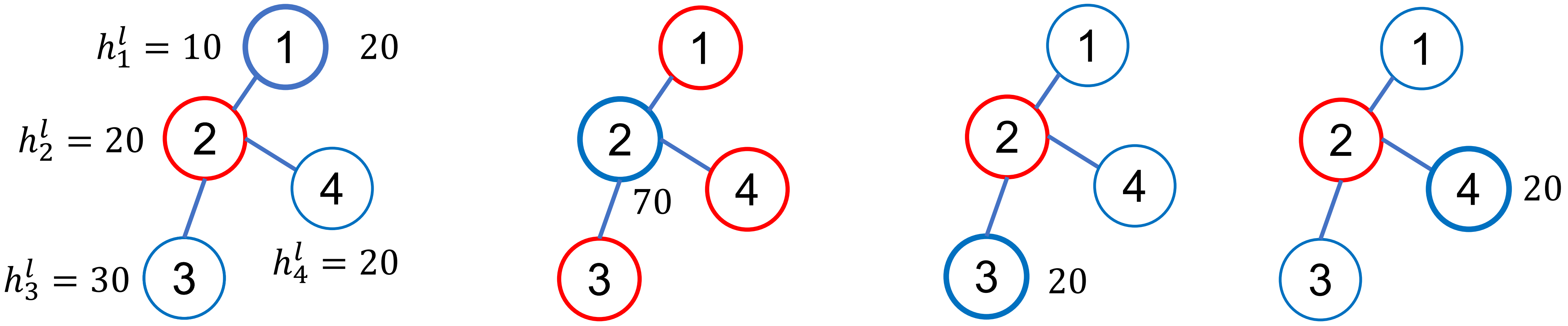
$\tilde{A}$: 3×3

$H^{(2)}$: $3\times\text{hidden\_size}$

W: $\text{hidden\_size}\times \text{num\_classes}$

可能有些 node(s) 的鄰居特別多

💡 GCN 使用相加來收集鄰居資訊



Source node: **1**
Neighbor node: **2**

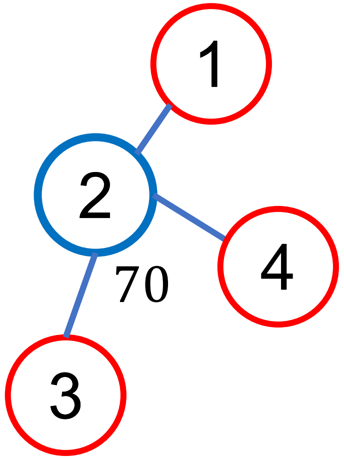
Source node: **2**
Neighbor node: **1, 3, 4**

Source node: **3**
Neighbor node: **2**

Source node: **4**
Neighbor node: **2**

⚠️ 此頁slide為求簡潔，每個node僅使用 1 個feature

GCN 對鄰接矩陣做標準化



$$\tilde{A}_{ij} \times \frac{1}{\sqrt{\mathrm{degree}(v_i)}\sqrt{\mathrm{degree}(v_j)}} H$$

⬅️ 標準化

Source node (i): **2**

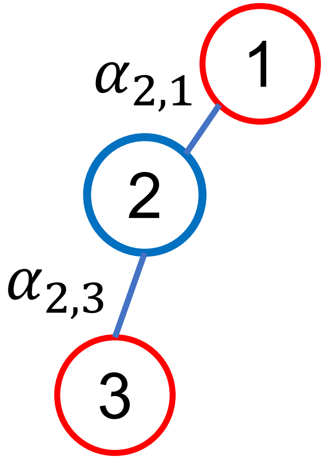
Neighbor node (j): **1, 3, 4**

Graph Attention Networks (GAT)

Petar Veličković, , Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, Yoshua Bengio. "Graph Attention Networks." International Conference on Learning Representations. 2018.

GAT 核心概念

$\alpha_{i,j}$ 代表 node j 對node i 的重要性



node1 對 node2 的影響有多大？

node3 對 node2 的影響有多大？

GCN 直接用相加來計算鄰居的重要性 ➡️
🤔 重要性可不可以讓模型自己學？

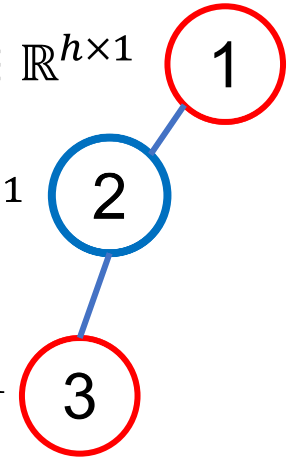
Attention Mechanism in GAT (1)

Step1: AGGREGATE

$\overrightarrow{h_{j1}^{l}} \in \mathbb{R}^{h\times 1}$

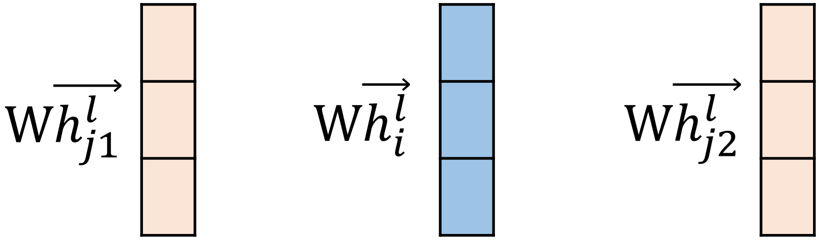
$\overrightarrow{h_{i}^{l}} \in \mathbb{R}^{h\times 1}$

$\overrightarrow{h_{j2}^{l}} \in \mathbb{R}^{h\times 1}$



每個節點都會經過線性

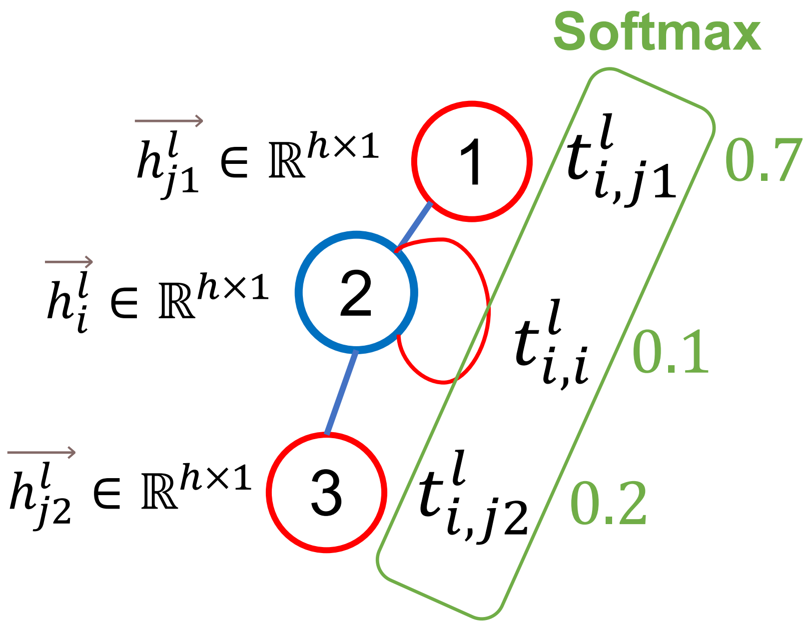
轉換 W ($W \in \mathbb{R}^{h\times h}$)

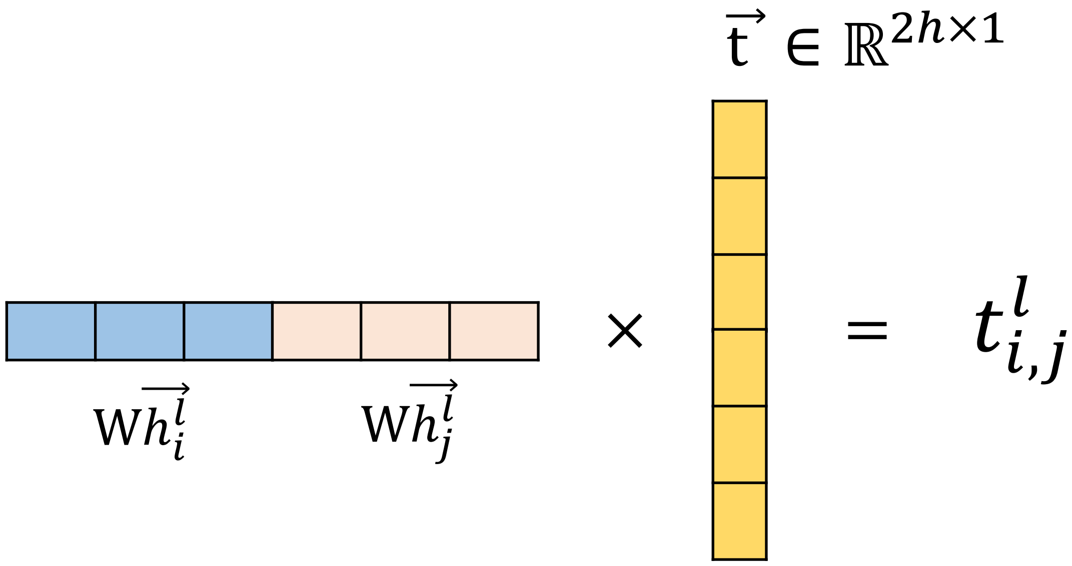


Attention Mechanism in GAT (2)

Step1: AGGREGATE

假設我們正在算 j 對 i 的重要性，$\vec{t}$ 是可以被訓練的參數

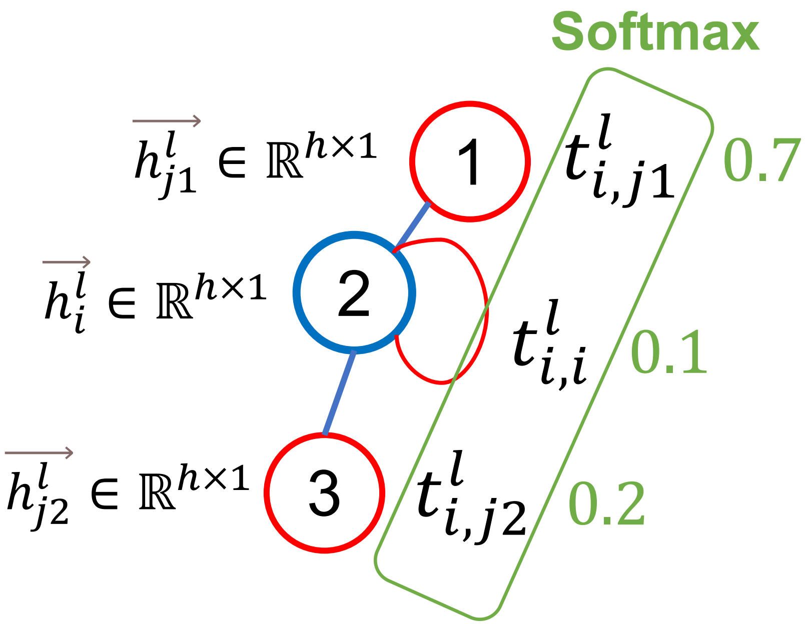




讓模型自己去學習哪些
鄰居比較重要 (attention)

GAT 更新節點狀態的方式 (UPDATE)

Step2: UPDATE



$$\overrightarrow{h_i^{l+1}} = 0.7 * \mathrm{W}\overrightarrow{h_{j1}^{l}} + 0.1 * \mathrm{W}\overrightarrow{h_{i}^{l}} + 0.2 * \mathrm{W}\overrightarrow{h_{j2}^{l}}$$

圖任務 Summary

假設有一張圖的 nodes 有3個，且每個 node 的 feature 數目為10個

GCN_layer2 $H^{(3)} = \tilde{A}H^{(2)}\color{red}{W}$ ➡ **Node Classification**

3×3

$3\times\text{hidden\_size}$

$\text{hidden\_size}\times\text{num\_classes}$

READOUT $h^{(G)} = MEAN(H^{(3)})$ ➡ **Graph Classification**

- 鄰居重要性差很多 (例如推薦系統)：優先嘗試 GAT
- 圖很大張：優先嘗試 GCN

延伸學習

- 李宏毅老師助教課
  - https://youtu.be/eybCCtNKwzA?si=bCBjHODKbrxGiGlz
  - https://youtu.be/M9ht8vsVEw8?si=qYy2ggGIuF8lEsE-
- 友善網誌
  - https://jacky-18008.medium.com/圖神經網路怎麼這麼棒-how-powerful-are-graph-neural-networks-42bbfd640e68
  - https://medium.com/data-science/graph-convolutional-networks-introduction-to-gnns-24b3f60d6c95
- 王佑中老師 slide
  - https://docs.google.com/presentation/d/1Xt_DThN7dd7w5Oay6RikgUhPx5wWw24xTQp7fQdEznA/edit?usp=sharing

Thank you!

Instructor: 林英嘉

✉️ yjlin@cgu.edu.tw

TA: 劉美辰

✉️ m1461014@cgu.edu.tw